

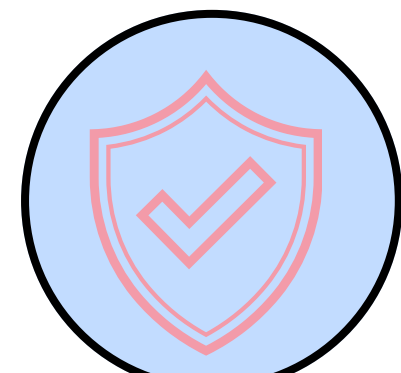
Background

During field trip departure, staff must manage attendance, forms, lunches, supervision, and wristband application all at once. Wristband application becomes a major staff burden.

Opportunity

Improve the wristband application process to reduce staff burden, increase efficiency, and enhance child experience.

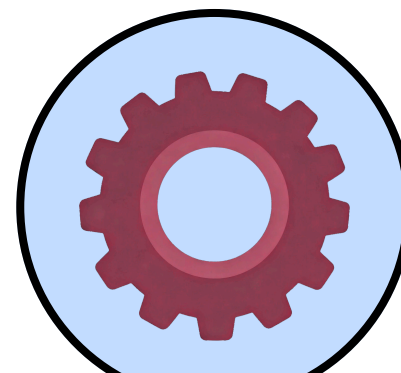
Design Objectives



Efficiency: faster application / lower setup time



Child Engagement: low-anxiety, approachable design



Low Staff Burden: minimal manual intervention



Safety: eliminate sharp hazards and reduce force on wrist

Next Steps

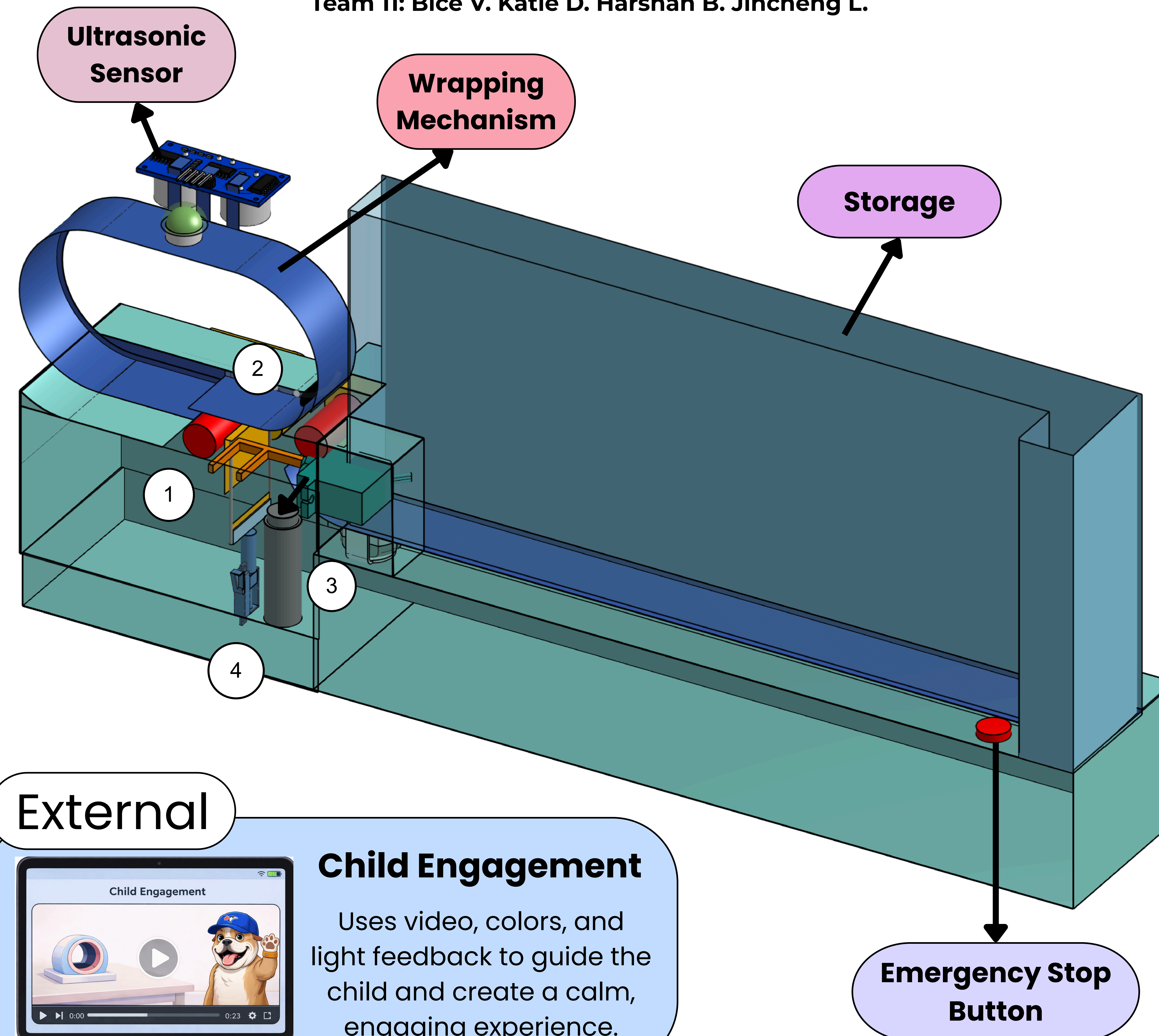
- Integrate full system
- Improve feeding mechanism
- Integrate Excess Management System
- Assess durability (long-term use)
- Select suitable materials
- Upgrade force sensor for adjustable control



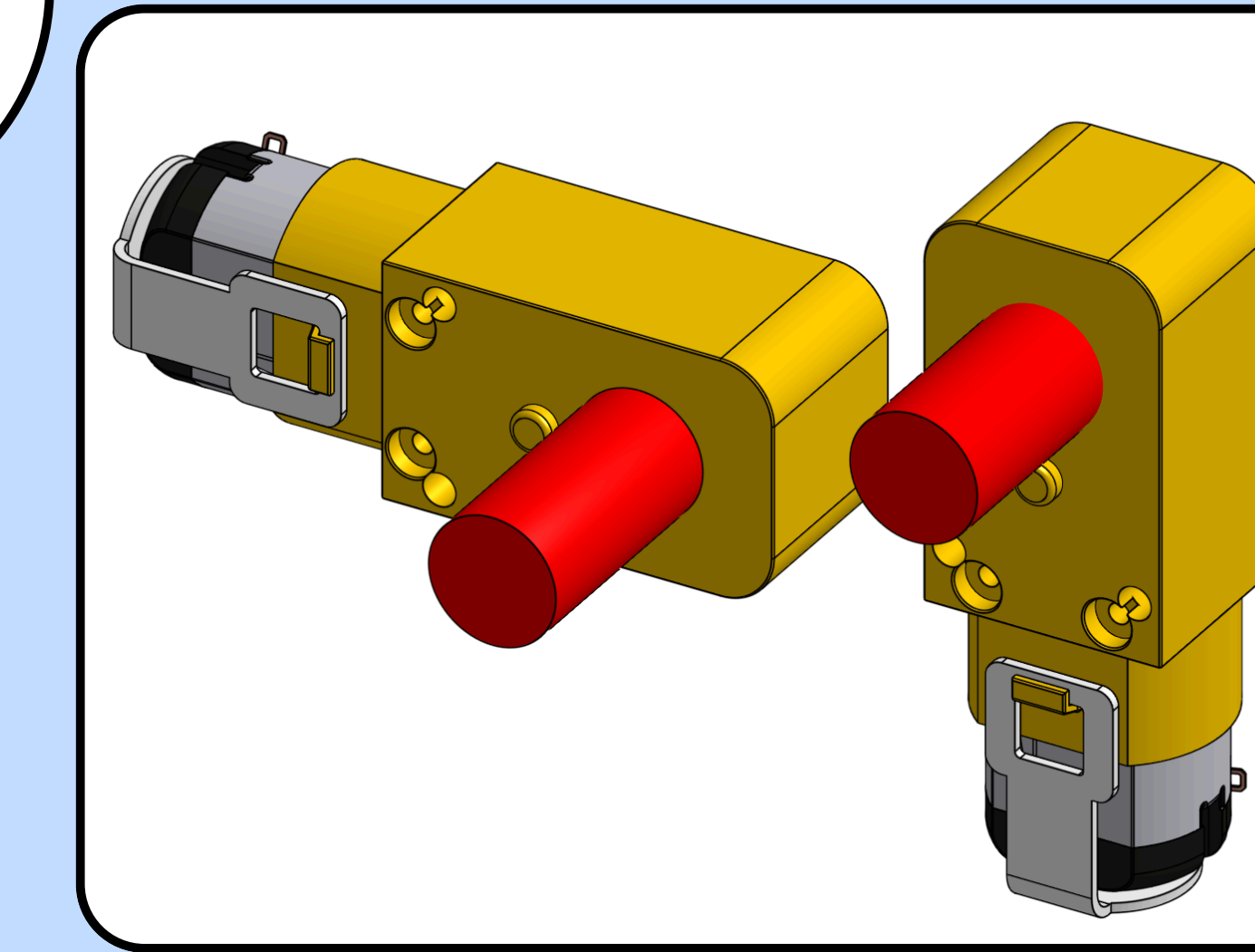
The SNAP BUDDY

AUTOMATED WRISTBAND APPLICATION SYSTEM

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1



Friction Rollers

Powered by the DC motors, these friction rollers respectively feed the bracelet through the mechanism.

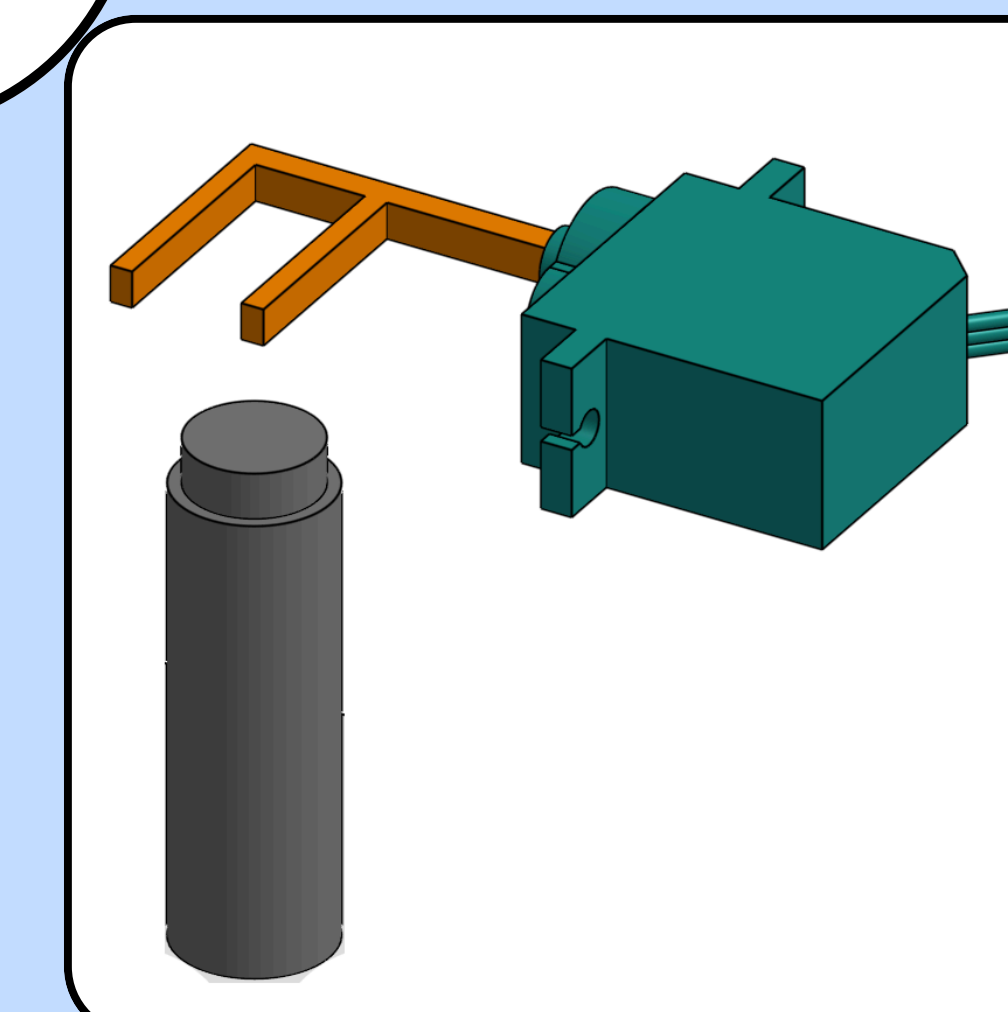
2



Thin Film Pressure Sensor

A sensor whose resistance changes with applied force. The Arduino reads this as a 0–1023 signal, which can be calibrated to estimate force.

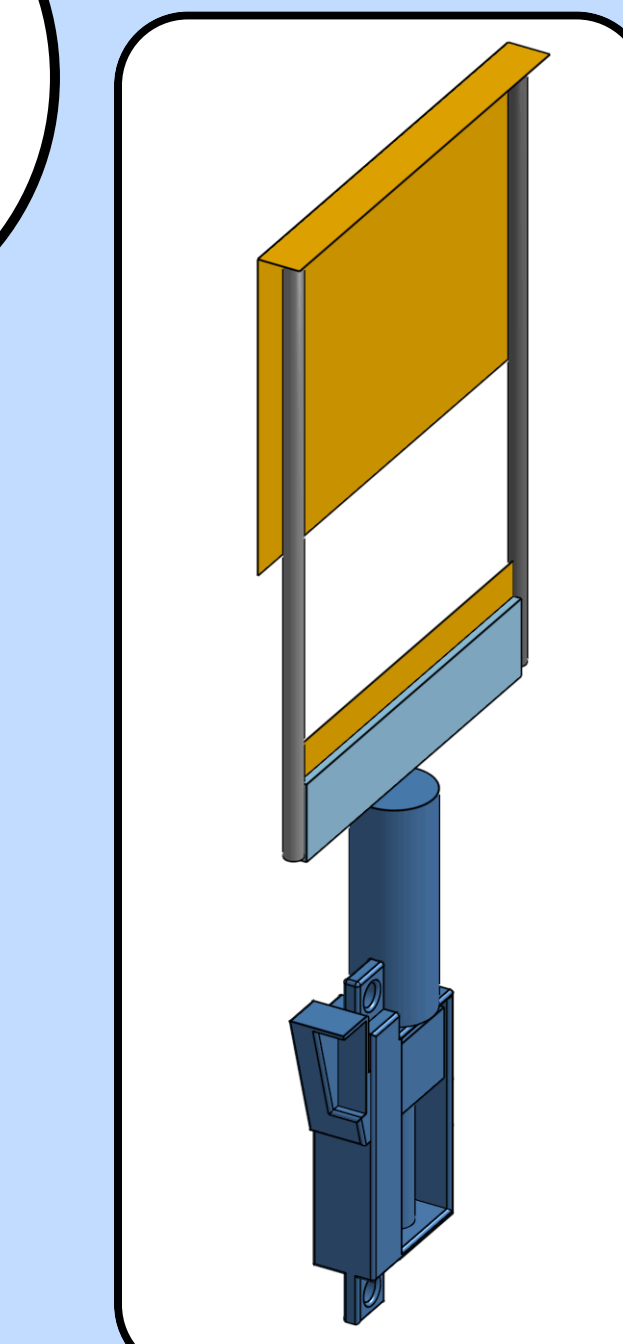
3



Buttoning Mechanism

A servo bends the bracelet tab into place, followed by a small piston that secures the button.

4



Cutting Mechanism

A linear actuator raises a razor blade that pushes the bracelet tail towards the button, then cuts the excess bracelet after the buttoning mechanism completes.